

Delta Exchange research — session record

Everything established on 2026-09-27, including the results that were negative and the ideas that were rejected. Written to stand alone if the machine it was produced on is gone. Generated 2026-09-27 10:31 UTC.

The one number that matters

Best strategy	Order block (SMC) + sentiment gate, ETHUSD 1d
Sharpe, 6y out-of-sample	0.94 — this is the number to quote
Quote instead of this	2.88 — the same strategy on 2 years of data
Sharpe at 2x costs	0.85
CAGR / max drawdown	+18.5% / -14.9%
Trades	129 over six years
ETH buy-and-hold, same window	+8.8%
BTC buy-and-hold, same window	+101% — do NOT use this strategy on BTC

Intraday: **0 of 32 configurations profitable**. 15m round-trip cost is 0.18%; the strategy's gross edge per trade is 0.0017%. Costs exceed the edge by 106x. On BTC, 10 of 16 configs were gross-positive (best +48.9%) and every one lost money net.

What was settled today

- **Sentiment gating helps, verified on six years.** Order block alone was Sharpe 0.60; gated to extreme Fear & Greed it reached 0.94. The 2.88 result was a two-year artefact.
- **Intraday does not work on this venue.** 32 configs, 15m bars, all negative. Measured, not assumed.
- **Time of day is real but insufficient.** 13:00-16:59 UTC holds 4 of 4 out of sample, yet it only halves fees by cutting trade count rather than adding edge.
- **Funding carry is real but too small.** Positive ~88% of days, 11-12% annualised, but a delta-neutral book doubles the cost and needs 60-365 days to clear its own fees.
- **ETH only.** Buy-and-hold beats this strategy on BTC by an order of magnitude.

Rejected, with reasons

- x Intraday scalping on Delta perps — 0/32 configs profitable, 106x cost drag
- x Funding carry at retail size — needs 60-365 day holds to clear fees
- x Time-window restriction alone — halves fees, does not add edge
- x TradingAgents repo — equities-only, not a perp execution engine
- x FVG, BOS, CHoCH patterns — 0/22 and 0/15 profitable configs
- x A YouTube candle-break strategy — 62 configs, 0 profitable
- x BTC 1d adx_trend_50_200 — Sharpe 1.36 on 21 trades, a meaningless sample
- x 1h timeframe — fees consume it, and sweeps time out on 2 cores

Bugs that produced fake results

Every one of these printed a spectacular number first. If a Sharpe above 3 appears, assume a bug until proven otherwise.

- ! Zero-return bars filtered out of the Sharpe denominator — **Sharpe 680**
- ! Order block tagged at its birth bar while reading 3 future candles — **87% win rate, 1643% CAGR**
- ! Pattern signals with no holding period, so fees were never charged
- ! Funding re-charged every bar because the window was bounded by entry_t not settled_t — 5% drag instead of 0.1%
- ! Epoch **seconds** read as milliseconds, collapsing dates to 1970 and printing **+8104% annualised** funding
- ! A DNS failure returning HTTP 0, which is numerically less than 400 and so passed a naive reachability check

Account

Balance	INR 35.01 (~\$0.41) — read balance_inr, not balance
API key	was in .env; emptied when Gemini/Groq keys were added
Trading permission	not enabled — /v2/positions returns 400
Live trading	impossible: \$0.41 against a ~\$10 minimum perp order
Shadow mode	cron 81116a955e5e, every 4h, no orders placed

The API key is not needed for any of the research above. Public candles and Binance funding history are enough. It is only required to read the account, and to place orders, which is not possible at this balance.

Published repos

delta-bot	Backtest engine + strategy. 11/11 engine tests pass. github.com/fsix7115-arch/delta-bot
freellm-probe	Which free LLM APIs actually work from your machine. Zero dependencies.
devcheck	Is the dev machine healthy, not just populated. Read-only, CI-safe, exit 1 when unhealthy.
Profile	github.com/fsix7115-arch — all three, MIT licensed

HANDOFF.md

Who the user is, account facts, the winning strategy, the five bugs, and the ranked next steps

COMPLETE HANDOFF — Delta Exchange Trading Research

=====

If you are a NEW agent reading this: everything you need is below.

Read STATE.md too — it has the live account status and current standings.

1. WHAT THE USER WANTS AND WHO THEY ARE

- **Language:** Hindi/English mix (Hinglish). Reply in Hinglish.
- **New to trading.** Does not know the markets. Explains things in simple words and expects plain answers, not jargon.
- **Limited time.** Their laptop may close at any moment, which kills the session. Everything is therefore persisted to disk, and a new session can resume in seconds.
- **The goal is NOT to get rich. The goal is that I "help them in trading" —** i.e. they want a partner doing the analysis while they look and learn.
- **Standing instructions** (they said these explicitly):
 1. Anything they send — check it, improve it, and if it is wrong, warn them and delete it.
 2. Find the best thing yourself; don't make them decide.
 3. Always tell them the ESTIMATED TIME before starting a long task, so they don't message in the middle and disrupt it.
 4. If they send something garbage, say so plainly. They respect that.
- They gave me a production Delta API key and a real (tiny) deposit. They repeatedly asked me to trade live. I declined to auto-trade. That boundary still stands and should be kept: ****suggest the trade, show the numbers, let them approve.****

2. THE ACCOUNT (verified against the live API)

- `.env` holds `DELTA_API_KEY`, `DELTA_API_SECRET`, `DELTA_BASE_URL`.
The key is a **PRODUCTION** key. `DELTA_BASE_URL=https://api.india.delta.exchange`.
- **Wallet: USD 0.4118 = INR 35.01.** That is all there is.
- **CRITICAL:** read `balance_inr` / `available_balance_inr`, NOT `balance`.
`balance` is the USD figure and reads as `~0` for an INR-only deposit. This bug made me wrongly report "no funds" four times before I found it.
- `/v2/positions` returns **400** — the key has `Read Data` but NOT `Trading` scope. Wallet, orders and fills all return 200.
- **Live trading is impossible anyway:** \$0.41 vs Delta's ~\$10 minimum perp order, and the strategy wants ~\$8 of margin at a 2% stop. Do not pretend otherwise.
- Product ids: **BTCUSD = 27**, **ETHUSD = 3136**. tick size 0.5.
- `.env` is `chmod 600` and is protected from being read back. Never print the secret. `check_api.py` and `account_dump.py` are read-only and safe to run.

3. THE STRATEGY THAT ACTUALLY WORKS

Order block (Smart Money Concepts), mechanised.

- Code: `strategies_smc.py` → `order_block_trade(df, impulse_atr=1.0, max_age=30, holdBars=8)`
- Venue/timeframe: **ETHUSD 4h on Delta**. Stop loss **2%**.

- Rule in plain words: a down-candle is tagged as an "order block" if the next 3 candles rally by more than $1 \times \text{ATR}$. When price later trades back into that candle's body, go long. Stop is 2%, target is $3 \times$ that.

Measured results, and which number to trust:

| Test | Sharpe | Trades | Verdict |

| Delta, 2y (2024-09 → 2026-09) | **1.01** | 376 | the headline number |

| Binance, 6y (2020 → 2026) | **0.60** | 78 | **the honest number** |

| 2x / 3x cost stress | survives | | has headroom |

| Monte Carlo (2y) | P(loss) 11.3% | | acceptable |

The 6-year Sharpe 0.60 is the one to quote. The 1.01 is one favourable 2-year window. Always show both.

- **ETH only.** On BTC, buy-and-hold (29.6% CAGR) beats the strategy (6.5%). On ETH, the strategy wins (14% vs 3.2%). Do not deploy this on BTC.
- Buy-and-hold on ETH over the same 6y window was only +3.2% CAGR, so the strategy's edge on ETH is genuine, just modest.
- Sentiment-gated variant (`ob_sent_gate` in `hybrid.py`) looked spectacular on 2y data (Sharpe 2.88, P(loss) 1.3%) but that is the max of 48 correlated trials on 41 trades. It has NOT been tested on 6y data. That test is the single most valuable next step.

4. THE FIVE BUGS I HIT — read these before trusting any number

Every one of these produced a spectacular, fake result. They are the reason this project took a long time, and the reason early numbers in my messages were wrong.

- **1. P&L double-counted.** `qty` is already in units, so P&L is `qty * (price - entry)`. Dividing by `entry` again gave Sharpe values like -40. Fixed in `engine.run`.
 - **2. Funding double-counted.** The funding window must be bounded by `settled_t`, not `entry_t`, or every settlement is re-charged on every bar. Gave a 5% drag instead of 0.1%. `test_funding_*` tests guard it.
 - **3. Zero-return bars were being filtered out of the volatility calculation.** A strategy that sits flat has *low* volatility; removing those bars shrinks the denominator and inflates Sharpe without bound. This produced **Sharpe 680**. `metrics()` no longer drops zeros, and reports `raw_sharpe` plus a `degenerate` flag.
 - **4. Lookahead leak in the order-block detector.** The block was tagged at its own birth bar while using the *next* three candles to confirm the impulse — reading a move that had not happened yet. Produced 87% win rate and 1643% CAGR. Fix: tag the block at bar `i+3`. `test_lookahead.py` guards it.
 - **5. Pattern signals were not held.** A pattern signal is an *entry event*, not a per-bar state. Without `holdBars` the engine opened and closed on the same bar and skipped the round-trip cost entirely.
- **Rule for any future strategy: if the Sharpe is above ~3, assume a bug until proven otherwise.****

5. DATA SOURCES (all free, no key except where noted)

- `data/{SYMBOL}_{15m,1h,4h,1d}.csv` — Delta's own candles, ~2 years max. Delta simply does not serve older data; this is a hard limit.
- `data/extended/` — Binance 6-year candles + **real funding-rate history** (6574 rows per asset, 8-hourly). Needed because Delta exposes no funding history and only 2 years of price.

- `data/sentiment/fear_greed.csv` — Alternative.me Fear & Greed, 3156 daily readings from 2018, free and keyless. A published composite (volatility 25%, momentum 25%, social 15%, surveys 15% paused, dominance 10%, Google Trends 10%). It is BTC-centric and market-wide, not per-coin.
- Sentiment is shifted in `_sentiment_at()` so a bar can only use a reading published before it. Keep that shift.
- **Live news headlines cannot be backtested honestly.** Today's headlines say nothing about 2022. If we want news-driven signals, it has to be a forward-looking shadow-mode feature, not a backtested edge.

6. WHAT WAS TESTED AND REJECTED — do not redo these

- **TradingAgents** (the repo the user sent): 22.9k lines, genuinely decent code with a real benchmark-alpha settlement loop. But it is equities-only (Yahoo Finance, SPY benchmark) and explicitly not a portfolio simulator. Its `settlement.py` idea is the one thing worth stealing later.
- **Stock Learners / the candle-break walkthrough**: mechanised in `strategies_youtube.py`, 62 configs, **0 profitable** with enough trades. The liquidity-sweep variant was flat 0.00 across all 32 configs.
- **ai4trade.ai**: an agent marketplace behind a login, not a strategy source. Could be a lead generator only.
- **FVG, BOS/CHoCH** in `strategies_smc.py`: 0/22 and 0/15 profitable.
- **1h timeframe**: structural failure, fees alone eat it. Sweeps time out repeatedly on 2 CPU cores.
- **Sentiment strategies on BTC**: -0.46 Sharpe. The F&G index is BTC-centric but does not predict BTC perps here; it worked on ETH.
- **BTC 1d `adx_trend_50_200`**: Sharpe 1.36 but only 21-23 trades. Meaningless sample. Rejected despite the headline number.

7. MACHINE AND OPERATIONAL NOTES

- **2 CPU cores only.** 4h/1d sweeps take ~6 min; 1h sweeps time out even at 2400s. Prefer 4h/1d and small grids.
- **Disk was 97% full.** `npm cache clean --force` freed 1.3GB; now ~70%.
- Always run `python3 test_engine.py` first (11/11 passing) before trusting results, and `test_lookahead.py` after touching any pattern detector.
- `bash ~/delta-bot/resume.sh` prints tests + current best + env status.
- The user's laptop closing kills the session. ****Shadow mode is scheduled on a cron job 81116a955e5e every 4h**** — it runs `shadow.py`, posts the current ETH 4h signal, and appends to `results/shadow_log.jsonl`. It places no orders.

8. SUGGESTED NEXT STEPS, IN ORDER

1. **Run the sentiment-gated hybrid on 6y Binance data.** This is the highest-value open question. `hybrid.py` has `ob_sent_gate`. If it holds ≥ 1.0 Sharpe on 6y, it becomes the primary strategy. Expect it to fall — that is a normal and useful outcome.
2. Test the winner on **BTC vs ETH separately** across more granular params to confirm the ETH-only finding rather than assuming it.
3. Add a **walk-forward parameter re-fit** (rolling in-sample window) instead of a single 55/45 split, so we can see how much the edge decays.
4. Only after 1-3: consider a **testnet run** (<https://cdn-ind.testnet.deltaex.org>, separate key). Not production.
5. Live trading requires: the user actually funding with a real amount (■5,000+), Trading permission enabled, a persistent host (this Cloud Shell

is ephemeral and would leave a position stranded if the session died), and per-trade approval.

9. TONE

The user is warm, calls me "bhai", and has limited time. Be direct, be honest about bad numbers, and never pad a reply. When a result is disappointing, say so plainly and show the next step rather than apologising at length. They have said "don't make me decide, find the best thing" — so bring a recommendation, not a menu, unless the decision is genuinely theirs (like risking real money).

INTRADAY.md

15m sweep, all 32 configs, and the 106x cost arithmetic

Intraday — the state of it

Date: 2026-09-27. The user asked specifically for intraday trading. This is what the data says, including the part that changes an earlier conclusion.

What changed

Earlier rounds dismissed intraday on reasoning alone: "fees eat it." That was a guess. Measuring it turned out more interesting than expected.

The dataset is good. Delta serves continuous 15m bars far deeper than its own documentation suggests:

| Symbol | Bars | Span |

| BTCUSD 15m | 96,168 | 2023-12-29 → 2026-09-26 (~1000 days) |

| ETHUSD 15m | 92,442 | 2024-02-06 → 2026-09-26 (~960 days) |

Found while checking, not assumed. The reader also needed fixing: the CSVs store epoch **seconds**, and reading them as milliseconds silently produced 1970 dates. Third occurrence of that same bug class in this project.

The fee hurdle

Delta perps, taker, no VIP tier:

- taker fee 0.05% per side → 0.10% round trip
- slippage 0.04% per side → 0.08% round trip
- **total friction 0.18% round trip**

Bar movement is large enough

| Lookback | BTC median | % of bars > 0.18% | ETH median | % > 0.18% |

| 1 bar (15m) | 0.104% | 29.4% | 0.150% | 43.1% |

| 4 bars (1h) | 0.198% | 53.3% | 0.287% | 65.3% |

| 16 bars (4h) | 0.401% | 72.6% | 0.595% | 81.6% |

| 96 bars (1d) | 1.195% | 88.8% | 1.732% | 93.0% |

About a third of 15m bars move more than the full round trip. So the earlier dismissal was wrong on the narrow claim: the hurdle is not impossible at this timeframe.

The backtest result

All 16 ETHUSD 15m configurations finished. **Every single one is negative.**

| impulse | hold | stop | gate | Sharpe | gross | net | trades |

| 0.6 | 8 | 0.40% | yes | -4.67 | +12.17% | **-96.17%** | 7,528 |

| 0.6 | 8 | 0.40% | no | -3.87 | +14.32% | -98.93% | 10,403 |

| 0.6 | 8 | 0.80% | yes | -4.19 | -0.19% | -92.63% | 5,738 |

| 1.0 | 8 | 0.40% | yes | -5.05 | -2.43% | -86.89% | 4,479 |

| 1.0 | 24 | 0.80% | no | -3.75 | -8.99% | -87.34% | 4,410 |

| 1.0 | 24 | 0.40% | yes | -4.01 | -6.34% | -89.82% | 4,914 |

Win rate 32-46% throughout, profit factor 0.81-0.86 — below 1.0 everywhere, which means the trades lose money even before fees are considered in the gross column for most configs.

The gross/net split is the actual finding

The first config is the informative one: **gross +12.17%, net -96.17%**.

`intraday_drag.py` measures this directly rather than inferring it:

- gross edge over 7,198 trades: +12.2%

- **break-even cost per trade: 0.0017%**
- venue cost per trade: **0.18%**
- costs exceed the edge by roughly **106x**

So the pattern is not broken. The logic finds the same structure it finds on the daily timeframe. It is simply too thin to be harvested ten thousand times a year instead of a hundred and thirty times.

This is the distinction that matters and a net-only report cannot make:

- gross healthy, net destroyed → venue problem, maybe fixable with execution
- gross also negative → no edge exists, nothing to fix

Here it is the first case, and the fix is not a better strategy. It is ****fewer trades****.

Full sweep: all 32 configurations, both symbols

results/intraday.csv, 16 configs each on ETHUSD and BTCUSD 15m.

| | ETHUSD | BTCUSD |

| Configs | 16 | 16 |

| Sharpe range | -5.05 to **-2.99** | -6.92 to **-4.07** |

| Positive Sharpe | **0 of 16** | **0 of 16** |

| Net return range | -99.1% to -78.0% | -97.9% to -78.8% |

| Positive net | **0 of 16** | **0 of 16** |

| Gross range | -25.1% to **+14.3%** | -16.6% to **+48.9%** |

| Positive gross | 3 of 16 | **10 of 16** |

| Trades | 3,228 - 10,403 | 3,026 - 9,301 |

Not one configuration out of 32 was profitable net. The best of the bad lot lost 78% of capital.

The BTC gross column is the more interesting one, and it was the last result in the run. ****10 of 16 BTC configs had a positive gross return, the best at +48.9%, while every one of them lost money net.**** ETH managed only 3 of 16.

That is a cleaner statement of the problem than the 106x figure: on BTC, this pattern does find profitable price movement. It captures roughly a quarter of capital gross across five to nine thousand trades — about 0.005% per trade — and then pays 0.18% per trade to the exchange. It is not that the strategy fails to find anything. It is that the venue takes far more than the strategy earns.

What the data says about trade count

| Config | Trades | Sharpe |

| ETHUSD 1d, 6y | 129 | **0.94** |

| ETHUSD 4h, 6y | 497 | 0.34 |

| ETHUSD 15m, 2.7y | 4,400 – 10,400 | -3.2 to -5.1 |

Sharpe falls monotonically as the timeframe gets faster. That is the finding: the same idea, harvested more often, loses to costs faster than it gains.

One detail worth noting: turning the sentiment gate **off** raised trade count from 7,528 to 10,403 and made net return worse (-98.93% vs -96.17%). More trades, more fees, no more edge. The gate was doing its job on the daily timeframe and here it only limits fee exposure.

Answer to "I want intraday"

Not on this venue, at this fee level, with this edge. The arithmetic is 106x against it, and that is not a tuning problem.

What would change the answer, in order of likelihood:

- 1. **Fewer, wider trades.** The one variant still worth running is a 1h or 4h hold computed on 15m bars, which cuts trade count by roughly an order of magnitude. The 24-bar hold tested here was not enough.
- 2. **Better fees.** A VIP tier or maker-rebate execution changes the 0.18% figure, and it is the only lever that scales.
- 3. **A genuinely different intraday edge.** Not this pattern. A structural product with no fees, or an order-flow signal that does not need to be harvested 10,000 times a year.

Absent one of those, ETHUSD 1d at Sharpe 0.94 remains the best thing found in this project, and it is the same strategy the user already has.

Reproducing

```
python3 intraday_fees.py # the fee hurdle and bar-move distribution
python3 sweep_intraday.py # the 16-config grid (~15 min on 2 cores)
python3 intraday_drag.py # the 106x arithmetic, measured not inferred
```

TIMEOFDAY.md

Hourly movement profile, out-of-sample validated, and why trading the peak window still loses

Time of day, and the three Stock Learners videos

Date: 2026-09-27

The videos

| ID | Title | Channel |

| wI9b968AvW8 | Best Trading Strategy For Beginners — Scalping Strategy | Stock Learners |

| XH0x0G1aAKw | Spot The Ultimate Move — Trap Trading Masterclass | Stock Learners |

| q0gfHJh526U | The Only Way For Success — Trading Psychology | Stock Learners |

None of the three have captions or transcripts, so their content could not be extracted. What follows about them is inferred from the titles and from what this project has already measured about the same channel, and is labelled as such.

The scalping video is the same channel and the same subject matter as `strategies_youtube.py`, which was mechanised from an earlier Stock Learners walkthrough: 62 configurations, best Sharpe 1.33 on 8 trades, median negative, zero configs profitable with enough trades to be meaningful. The intraday work below independently confirms that conclusion on real 15m data with real costs.

What the titles imply, and what was actually tested:

- **Scalping** → tested. Negative. See the 16-config 15m sweep and the 106x cost arithmetic in `INTRADAY.md`.
- **Trap Trading Masterclass** → the mechanics are order blocks and liquidity sweeps, which are already implemented in `strategies_smc.py`. Order block is the winning strategy. FVG, BOS and CHoCH were tested and produced 0 of 22 and 0 of 15 profitable configs.
- **Trading Psychology** → not testable by backtest. Noted and set aside.

The time-of-day question

The request was to find when the large players are active, on the theory that activity means opportunity. Measured on ~96,000 fifteen-minute bars per symbol over roughly 2.7 years, all times UTC.

The pattern is real and it survives out of sample

| Hour (UTC) | BTC median move | ETH median move |

| 14:00 | 0.165% (1.59x) | 0.228% (1.53x) |

| 15:00 | 0.148% (1.43x) | 0.207% (1.39x) |

| 16:00 | 0.134% (1.29x) | 0.188% (1.26x) |

| 13:00 | 0.131% (1.26x) | 0.187% (1.25x) |

Weakest hours are 04:00-06:00 and 10:00-11:00 UTC, at 0.075-0.11%.

The dataset was split in half. The four best hours in the first half were

[13, 14, 15, 16] and in the second half they were also [13, 14, 15, 16].

Four of four overlap. This is not a coincidence found in one period.

14:00-17:00 UTC is the European/US overlap, which is where institutional activity and liquidity peak. The folklore was right about that much.

But does it help to trade it?

No. This is where the honest answer is less satisfying than the pattern.

| Symbol | Variant | Sharpe | Gross | Net | Trades |

| ETH | all hours | -5.05 | -2.43% | -86.89% | 4,479 |

| ETH | **window 13-16 only** | -4.25 | -3.91% | **-45.37%** | 1,318 |

ETH	outside window only	-5.55	+0.40%	-79.26%	3,497
BTC	all hours	-6.92	-8.22%	-86.05%	4,229
BTC	**window 13-16 only**	-5.76	-6.36%	**-46.55%**	1,267
BTC	outside window only	-7.60	-1.68%	-77.65%	3,309

Restricting to the active window roughly **halved the loss** on both symbols.

That is a real improvement and it is also still deeply negative.

Two things to read here:

- 1. **Trade count fell from ~4,400 to ~1,300, and the loss halved.** Almost exactly proportionally. The gain is from trading less, not from trading better. Fees are per trade; cutting trades by two-thirds cut the bill by two-thirds. The window did not add edge, it removed cost exposure.
- 2. **Gross return went DOWN inside the window** on both symbols (-2.43% → -3.91% on ETH, -8.22% → -6.36% on BTC, mixed). A genuine volatility edge would show gross improving. It did not. What improved was net, and only because fewer fees were paid.

The sanity check also came out in the expected direction: trading the **dead** hours was worse than trading the peak hours (-5.55 vs -4.25 Sharpe on ETH). So the time-of-day story is not inverted, it is simply insufficient.

The conclusion

Big players are active, liquidity peaks, and price moves 1.3-1.6x more at

- 14:00-17:00 UTC than at 05:00. All of that is measured and holds out of sample.

It does not create an edge. The cost per trade on Delta is 0.18% and the strategy's gross edge per trade is 0.0017%, so the gap is 106x. A window that tells you **when** price moves does nothing about the fact that a 0.18% round trip needs a 0.18% move just to break even. ETH's best hour clears that hurdle on the median bar; BTC's does not, at 0.165% against a 0.18% cost.

If the user wants to trade the active window anyway, the correct configuration is 13:00-16:59 UTC, and it should be paired with the daily timeframe logic, not with 15m scalping. Running the daily strategy on 15m bars and entering only inside the window is the one combination here with a plausible mechanism, and it is worth a test.

Files

```
python3 tod_profile.py # hourly movement profile + OOS validation (~2 min)
python3 tod_backtest.py # window-restricted backtest (~10 min on 2 cores)
```

Results in results/tod_window.csv.

FUNDING.md

Delta-neutral funding carry tested on six years of real funding rates, and why it is not deployable

Research log — funding carry and the direction question

Date: 2026-09-27

The request

"I want to make money whether the market goes up or down. A single trade should be able to profit. Just tell me buy or sell."

This is worth taking seriously, because the delta-neutral carry trade is genuinely designed for exactly that: it collects funding regardless of direction. So it was tested properly rather than dismissed.

Result: real, but too small to trade retail

Six years of real funding data, 6574 settlements per asset, 2020-09 to 2026-09.

| | BTCUSD | ETHUSD |

| Days with positive funding | 88.3% | 86.8% |

| Mean funding per day | +3.04 bp | +3.36 bp |

| Annualised, perp leg only | +11.1% | +12.3% |

| Median 30-day hold, gross | +0.53% | +0.50% |

The long-run bias is real. Funding is positive on roughly 7 days in 8, which is the whole reason the trade exists.

Costs decide it. A round trip costs 0.10%, and fees are paid **once** while funding accrues every 8 hours, so the cost amortises with the hold:

| Hold | BTC p5 net | ETH p5 net |

| 7d | -0.13% | -0.16% |

| 30d | -0.15% | -0.22% |

| 90d | +0.32% | -0.52% |

| 180d | +0.83% | -0.72% |

| 365d | +3.42% | +1.65% |

p5 is the worst 1-in-20 window. BTC needs a 60-day hold before the unlucky case clears its own fees. ETH needs a full year. The median return over that year is about 7%, on capital locked up for twelve months.

Why this is not the answer

Three things, in order of how much they matter:

- **1. These are perp-leg-only figures.** A genuine delta-neutral book is short perp + long spot. The spot leg pays its own round-trip fee, which roughly doubles the cost. Every number above should be cut about in half to become a real trade. Delta serves no public spot candles, so this could not be simulated honestly and was not simulated at all.
- **2. A short-perp-only version is not direction-neutral.** It profits when price falls. Anyone running it as shown is making a directional bet and calling it carry. The "88% of days positive" figure is a statement about funding, not about price.
- **3. It needs capital to sit still for months.** The entire edge is a fraction of a percent per month. At INR 35 (~\$0.41) there is no position to hold, and even at INR 50,000 the absolute numbers would be a few hundred rupees a month, for a year of locked capital.

This is why exchanges market "passive income from your idle funds" rather than telling you to run the trade yourself. The yield is real and it is small.

Two bugs found while producing these numbers

Worth recording, because both produced spectacular results first:

- 1. **Timestamp unit.** The funding CSVs store epoch **seconds**. Parsing with `unit="ms"` gave 1970 dates, which collapsed the daily grouping and printed "+8104% annualised". `load_funding` now detects the unit.
- 2. **Column name.** The rate column is `rate`, not `funding_rate`. The loader now probes a list of candidate names.

The 8104% number appeared in output before it was caught. A headline that extreme is a bug until proven otherwise.

What this changes about the plan

Nothing about the primary strategy. The order-block plus sentiment-gate result (Sharpe 0.94 on six years, ETHUSD 1d) remains the best thing found, and it is direction-aware by design: it takes long or short depending on what the setup shows. The user's framing of "I just want to know buy or sell" is compatible with that, and the strategy already answers that question on every bar.

The funding carry is documented here so it does not get re-investigated. It is not deployable at any realistic account size.

SHADOW.md

What shadow mode is, and how the log gets scored

Shadow mode — what it is and what it is not

The short version

Shadow mode runs the winning strategy on live Delta candles and logs what it would have done. It places no orders and cannot, because the account holds INR 35.01 (~\$0.41) against a ~\$10 minimum perp order.

What it is for: finding out whether a strategy that looked good on six years of backtest behaves the same way on prices nobody has seen yet.

The files

| File | Role |

| `shadow.py` | Fetches live candles, prints the current signal, appends to the log |

| `shadow_score.py` | Replays the logged signals forward and scores them |

Run both:

```
python3 shadow.py # what is the signal right now
python3 shadow_score.py # were the logged signals any good
```

A cron job (81116a955e5e) runs both every 4 hours and posts the result here.

Why scoring is separate

A signal log on its own proves nothing. A system that is always long produces a signal every single bar and makes no money. The only question worth asking is whether following the log would have beaten not following it.

`shadow_score.py` replays each logged signal forward against the actual candles using the same 2% stop and 8-bar hold as the backtest, then reports the same shape of number. It also runs an **always-long baseline** with identical exit rules, because "did better than doing nothing" is a much lower bar than "is good" and it is worth knowing which one was cleared.

No lookahead: a signal logged at bar T is applied from bar T+1's open, matching `engine.run`.

Current state

3 log entries, 1 completed trade, +1.08%. **That is not a result.** Six entries a day means 30 entries takes five days, and 30 entries is still thin. Nothing should be concluded before then.

The backtest number for the same strategy is Sharpe 0.94 over six years on ETHUSD 1d. Shadow mode runs ETHUSD 4h, which scored lower (0.34) on the same six-year test — so the 4h shadow log is running the weaker of the two configs.

That is a deliberate choice: 4h gives six checks a day instead of one, which accumulates evidence faster, and the score is reported honestly either way.

What would change the verdict

- Shadow results diverging sharply from backtest, in either direction
- A drawdown materially worse than the backtest's -14.9%
- Signal frequency collapsing or doubling

Any of those and the strategy goes back to the drawing board regardless of what the backtest says.

STATE.md

Live account status and current standings

PROJECT STATE — read this first in any new session.

What this is

Delta Exchange India BTC/ETH perpetual futures research. Located at `~/delta-bot`.

Phase: research/backtest only. Live trading NOT started, NOT approved.

Account status (verified against the live API)

- Production key works. Wallet **USD 0.4118 / INR 35.01**.
- `account_dump.py` reads `balance_inr` — the plain balance field is USD and reads as `~0` for an INR-only deposit. That caused a wrong "no funds" report.
- `/v2/positions` still 400s: the key has Read Data but **not Trading** scope.
- **Live trading is impossible right now**: \$0.41 available vs Delta's ~\$10 minimum perp order, and the strategy needs ~\$8 margin at a 2% stop.
- Shadow mode (`shadow.py`) is the active setup: logs signals, places no orders. Cron job `81116a955e5e` runs it every 4h and posts the signal here.
- `~/delta-bot/results/shadow_log.jsonl` accumulates every observation.

Current best result

Winner: order-block strategy on ETHUSD 4h (Delta data, 2y)

- `strategies_smc.order_block_trade(impulse_atr=1.0, max_age=30, hold_bars=8)`, stop 2%
- OOS Sharpe **1.01**, 376 trades, CAGR +21%, max DD -16%, win rate 49%, PF 1.20
- Buy-and-hold ETH same window: **-21.6%** => alpha **+43%/yr**
- Monte Carlo (2000 bootstraps): P(loss) **12.2%**, p5 -12.6%, worst DD -31%
- This is the only config found with P(loss) under 15%.

Previous leader, now demoted: `adx\_trend\_20\_50` BTC 4h (6y Binance)

- OOS Sharpe 0.79, 70 trades, CAGR +15.5%, P(loss) 28.2% — MARGINAL
- ETH 4h version of this FAILED on Delta data (Sharpe -0.10); the 6y Binance result did not transfer. Do not re-quote the 6y numbers as if they were live-venue results.

Rejected: BTC/ETH 1d `adx_trend_50_200` (21-23 trades, meaningless sample).

Stock Learners candle-break: 0/62 configs profitable with enough trades.

Files

- `fetch_data.py` — Delta 2y candles (no key needed)
- `fetch_extended.py` — Binance 6y candles + real funding history
- `engine.py` — backtest engine (fees, slippage, stops, funding, no-lookahead)
- `strategies.py` — 20 signal generators
- `sweep.py` — walk-forward sweep, `SOURCE/RESOLUTIONS` env vars
- `test_engine.py` — **11/11 passing**, run this before trusting any result
- `.env.example` — template for API keys (user fills, `chmod 600`)
- `results/sweep_6y.csv` — 155 configs, 31 passed the bar

Hard-won lessons (do not relearn these)

1. **Delta only serves ~2 years of candles**. 2020 crash and 2022 bear market are NOT available from Delta. Binance is the only source for 6y — but a Binance result does NOT transfer automatically (proved: ETH 4h `adx_trend` went from Sharpe 1.12 on Binance to -0.10 on Delta).
2. **`run()` must use `settled\_t`, not `entry\_t`, to bound the funding window**. Using `entry_t` re-counts every settlement each bar and produced a 5% drag instead of 0.1%. A test now catches this.

- 3. **qty is in units** — P&L is ``qty * (price - entry)``. Dividing by entry again double-counts and produced absurd Sharpe values like -40.
- 4. ``metrics()`` **must NOT drop zero-return bars**. Filtering `r[r != 0]` removes the flat periods, shrinks the volatility denominator and inflates Sharpe without bound (produced Sharpe 680 on an 8-bar-hold strategy). `metrics()` now reports `raw_sharpe` and a `degenerate` flag for same-bar open/close artifacts (`avgBars < 1.5`).
- 5. ****Pattern detectors must tag the bar the pattern is KNOWN, not the bar it started.**** An order block tagged at its own birth bar reads the impulse that has not printed yet — this leaked future data and produced Sharpe 60 with 87% win rate. `test_lookahead.py` guards it now. Every stateful pattern needs a `holdBars` so the engine does not open and close on the same bar.
- 6. Only 2 CPU cores. 1h sweeps time out repeatedly; 4h/1d finish in ~6 min.
- 7. Disk was 97% full; `npm cache clean --force` freed 1.3GB. Now ~69%.
- 8. Long sweeps need `timeout 2400`; the default 850s kills them.
- 9. `.env` must use exactly `DELTA_API_KEY` / `DELTA_API_SECRET`. The key in it is a PRODUCTION key (testnet + prod-IN 401, only prod-IN 200). Balance 0. Wallet reads work; `/v2/positions` 400s because the key lacks Trading scope.

Next steps (in order)

- 1. ~1h timeframe sweep — running/complete, check `results/sweep1h.log`
- 2. **Delta cross-validation** — re-run the top strategy on Delta's own 2y data with real Delta fees. THIS MATTERS: all current numbers are Binance.
- 3. **Monte Carlo** — shuffle trade order, 1000 runs, check the strategy survives
- 4. Testnet paper trading (4-6 weeks) BEFORE any real money
- 5. Only then consider live, and only with per-trade approval

Quick resume

New session: just say "Resume delta-bot", or run `bash ~/delta-bot/resume.sh`.

Everything needed is on disk — no need to re-explain anything.

The user has limited time (laptop may close), so keep state disk-persisted.

Read `HANDOFF.md` first if you are a new agent. It contains the full brief:

who the user is, the account facts (including the `balance_inr` trap), the working strategy, the five bugs that produced fake results, what was already tested and rejected, machine limits, and the ranked next steps.

What the user has NOT approved

- Giving the API key (they asked for a `.env` template instead — done)
- Live trading
- Any real-money order

Standing user instructions

- Check anything they send, improve it, warn and delete it if it's not usable
- Estimate time BEFORE starting any long task so they aren't disturbed
- They have limited time; leave state on disk so a new session resumes instantly